MATH 440/540-01 Applied Math I: PDE's (Fall 2024)

Lynn Schreyer (she/her/hers) MWF 12:10-1:00pm; EME B46

Class number: 440-01: 04503 540-01: 04504
Credit Hours: 3
Instructor Information:

Office: Neill Hall 225
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Email: Lynn.Schreyer@wsu.edu
Web Page: math.wsu.edu/faculty/lschreyer (first letter is "ell", all lower case)
Student Hours:

MWF after class.
Wed 5:45-6:30pm Zoom
Meeting ID: 983 7218 9111
Passcode: 236278
Thurs 5:30-6:45pm MLC, Cleveland Hall 130

Textbooks: There are two textbooks for this course. The primary textbook is:

Introduction to Partial Differential Equations
E. C. Zachmanoglou and D. W. Thoe, 1986, Dover
ISBN-13: 978-0-486-65251-1
ISBN-10: 0-486-65251-3

and the optional reference book is

An Introduction to Partial Differential Equations
By Yehuda Pinchover and Jacob Rubinstein
ISBN-13: 978-0-521-61323-1 (paperback)
ISBN-10: 0-521-61323-X (paperback)
Available for free at:
https://www.academia.edu/33064706/Yehuda_Pinchover_Jacob_Rubinstein_An_Introduction_to_Partial_Differential_Equations

The Zachmanoglou and Thoe textbook is a Dover book (on the order of $15), and is a really great textbook for bridging PDE's at the undergraduate and graduate level. It covers some theory, without needing graduate level mathematics. The disadvantage is that it is old – no pretty color pictures, less motivation, and it could be written more clearly. The Pinchover and Rubinstein book is free, covers mostly the same material in Chapters 1-6, and 7 and motivates with physical examples.

Topics: Math 440/540 provides an applied introduction to the subject of partial differential equations (PDEs). Topics include (but are not limited to) first-order PDE's (linear and nonlinear) solved by method of characteristics, second-order PDE's (Laplace, heat equation, and wave equation) solved by method of separation of variables, Green's functions, Bessel functions, and the interplay between the mathematics and the physics.

Prerequisite: MATH 315 (Ordinary Differential Equations). It will be assumed that you are familiar with material provided in a basic 1-semester course of ordinary differential equations. The first homework assignment will cover most of the material you will be expected to know. There is not time during class to cover the prerequisite material, so please review.

What Are the Goals of this Course?

A student who passes this course will be able to:

1. Use standard techniques for solving LaPlace's, Heat, and Wave equations
2. Find the Fourier transform given an appropriate function.
3. Use the Method of Characteristics to solve a first-order quasi-linear PDE
4. Understand a physical problem for which canonical PDE's are appropriate models.
5. Use theorems about existence and uniqueness to determine if a PDE problem is well-posed.
6. Write up solutions to mathematical problems so that a classmate understands the approach and method for the problem.
7. For 540 students, students will be able to learn PDE material independently and explain the material in writing to classmates.

The assessment of these learning outcomes will be based on the following:

MATH 440:
The course grade will be based on

- weekly assignments (homework) (assesses outcomes 1-6)
- three in-class exams (assesses outcomes 1-5)
- a comprehensive final exam (assesses outcomes 1-5).

The point distribution will be weighted as follows:

Homework:	20%
Test 1	15%
Test 2	15%
Test 3	15%
Final:	35%
Total:	100%

MATH 540:
The course grade will be based on

- weekly assignments (homework) (assesses outcomes 1-6)
- three in-class exams (assesses outcomes 1-5)
- *a written project* (assesses outcomes 7)
- a comprehensive final exam (assesses outcomes 1-5)

The point distribution will be weighted as follows:

Homework:	15%

Test 1	15%
Test 2	15%
Test 3	15%
Written Project	15%
Final:	25%
Total:	100%

The written project will be an individual project on a topic of the student's choice but the topic must be approved by the instructor. As we go through the course, take note of topics that we skip in the textbook, or a problem you find particularly interesting. Alternately, if there is a PDE or integro-differential equation in your research area that you would like to know more about, this is your opportunity! The paper should be typed, 1.5 spaced, 12-15 pages, and 12 point sized font. More information about the written project will be provided at a later date.

Your final grade for both 440 and 540 will be determined on a straight scale:

90-91.9% A-	92-100% A	(A+ can not be given)
80-81.9% B-	82-87.9% B	88.0-89.9% B+
	70.0-77.9% C	78.0-79.9% C+
	60.0-69.9% D	
	Below 60.0% F	

Exam Schedule:

Test 1: Fri. Feb. 10
Test 2: Fri. March 10
Test 3: Fri. April 21
Final Exam: To be determined. Week of May 1-5.

Assignments:

Assignments for each week are due on the following Thursday by midnight (technically 11:58pm via Crowdmark). Assignments may be turned in up to 24 hours late with a 33% penalty. Assignments should be neat, well-organized, and submitted as laid out in Crowdmark. It will help if you start each new problem on a separate sheet of paper. If you have technical difficulties or turn in your homework an hour late you will not be penalized for being late.

Your lowest two homework grades will not be counted toward your homework grade. Assignments copied from a solutions manual or outside source or two students submitting duplicate homework solutions without citation may be given a

grade of zero. *Assignments turned in 24 hours after the due date and time will not be accepted,* except under special circumstances such as being out for a week or longer. Note that having too many assignments/projects due in another class or being ill for 4 days or less will not be considered a special circumstance. Please plan accordingly, and assume you will be very busy or will get sick when the last two assignments are due.

To be eligible to obtain a perfect score on your homework, a summary of the problem and problem goals must be provided, and an explanation of why you are doing what you are doing (enough so that a struggling classmate can understand how to do the problem without referring to external resources) must be provided.

How Will You Meet the Goals of this Course?

You have many resources, please use them!

- *Attend class,* where much of the material will be explained and motivated.
- *Work with classmates* on homeworks.
- *Attend student hours* of your instructor or another instructor and get specific questions answered. Also this is where you can meet classmates.
- Homework is where you explain how you are thinking, and then get feedback on your thinking process. *Start your homework early* to allow for ideas to percolate.

Math Learning Center:

Successful students make use of available resources, so don't struggle when help is just a few steps away! We want you to succeed, we're here for you, and we have FREE tutoring available in the Math Learning Center (Cleveland 130).

The MLC is located in Cleveland Hall 130. Tutoring begins August 19th. We are open

Mon-Thurs: 10am-9pm
Fri: 10am-5pm
Sun: 4-9pm

Zoom tutoring will also be available. Information can be found on our website: http://www.math.wsu.edu/studyhalls/welcome.php

To be successful:

It is expected that students will attend almost every class, and will spend 6-9 hours per week outside of class reviewing notes and working on homework and/or reviewing for exams. Students who are rusty with prerequisite material may spend 9 or more hours per week on this class. If you find yourself in this upper range, please attend student hours and get individual help – it will speed up your learning process! Do not wait until the day before homework is due to start working on it – leave time to get your questions answered!

Attendance

Students should make all reasonable efforts to attend all class meetings. However, in the event a student is unable to attend a class (*such as if a student is ill)*, it is the responsibility of the student to inform the instructor as soon as possible, explain the reason for the absence (and provide documentation, if appropriate), and make up class work missed within a reasonable amount of time, if allowed. Course lectures will be recorded and made available upon request (up to technological issues). Attending class regularly and working through all homework problems are highly correlated with higher grades. Attendance will be taken randomly and will be used to determine grades that are borderline.

Academic Integrity

I encourage you to work with classmates on assignments. However, each student must turn in original work. No plagiarism (copying without referencing source) will be accepted either from a classmate or an online resource.

Academic integrity is the cornerstone for higher education. As such, all members of the university community share responsibility for maintaining and promoting the principles of integrity in all activities, including academic integrity and honest scholarship. Academic integrity will be strongly enforced in this course. Studetns who violate WSU's Academic Integrity Policy (identified in Washington Administrative Code (WAC) 504-26-010(3) and -404) may receive an F as a final grade in this course, will not have the option to withdraw from the course pending an appeal, and will be reported to the Center for Community Standards. If you wish to appeal a faculty member's decision relating to academic integrity, please use the form available at communitystandards.wsu.edu. Make sure you submit your appeal within 21 calendar days of the instructor's decision.

Cheating includes, but is not limited to, plagiarism and unauthorized collaboration as defined in the Standards of Conduct for Students, WAC 504-26-010(3). Read and understand all of the definitions of cheating: http://app.leg.wa.gov/WAC/default.aspx?cite=504-26-010. If you have any questions about what is and is not allowed, ask your course instructor.

University Syllabus Statement and Link

Students are responsible for reading and understanding all university-wide policies and resources pertaining to all courses (for instance: accommodations, care resources, policies on discrimination or harassment), which can be found in the university syllabus.

Class Recordings

We will use Zoom for virtual meetings and recordings in this course. The recordings of class will only be provided upon request. Our use of such technology is governed by FERPA, WSU's Electronic Communication Policy

(EP4: https://policies.wsu.edu/prf/index/manuals/executive-policy-manual-contents/ep4-electronic-communication-policy/) and WSU's Student Code of Conduct (https://communitystandards.wsu.edu). A record of all meetings and recordings is kept and stored by WSU, in accordance with WSU's Executive Policies and FERPA. Your instructor will not share recordings of your class activities outside of course participants, which includes your fellow students. *You may not share recordings with individuals outside of this course. Doing so may result in disciplinary action.*

Important Dates:

Sunday Aug. 25: Last day students may add a course on-line. Classes added after this date require appropriate signatures.

Friday Sept. 6: Last day a student may change enrollment from letter grade to pass/fail or from credit to audit or audit to credit.

Tuesday Sept. 17: Deadline for dropping a course without record. Course withdrawals after this date are recorded on the student's transcript.

Friday Nov. 15: Deadline for undergraduate students to withdraw from a course. Withdrawals do not reduce tuition charges.

Friday Dec. 6: Deadline to change from pass/fail to a letter grade.

Wed. Dec. 18: Final Grades available.

For a complete list of dates, please see: http://registrar.wsu.edu/academic-calendar/

Tentative Schedule

Week of	Topics	Notes
Aug. 19	Ch 1: Sections 1-3: Sets and Functions, Surfaces, Implicit Function Theorem, Curves and their tangents	
Aug. 26	Ch 1: Section 4 ODE Solution of PDE; Classification of First-order PDE; Definition of Domain; Method of Characteristics	
Sept. 2	Method of Characteristics (continued) Ch 3 Section 5: Conservation Law	No class on Mon. Sept. 2 (Labor Day)
Sept. 9	Ch 3 Section 6: Traffic flow problem Ch 4 Taylor Series, Cauchy-Kovalevsky Theorem	
Sept. 16	Ch 5: S 7: Classification and Canonical forms of second-order PDE's	**Test 1:** Fri. Sept. 20
Sept. 23	Ch 6 Sections 1-5: Green's Identities and Introduction to second-order PDE's Start Ch 7: Laplace Eqn and Fourier Transform	
Sept. 30	Ch 7: (cont)	
Oct. 7	Ch 7 (continued)	
Oct. 14	Ch 7 (continued)	**Test 2:** Fri Oct. 18
Oct. 21	Ch 8: Wave Eqn	
Nov. 28	Ch 8: Wave Eqn	
Nov. 4	Ch 8: Wave Eqn	
Nov. 11	Start Ch9: Heat Eqn	No class on Mon. Nov. 11 (Veteran's Day)
Nov. 18	Ch 9: Heat Eqn	**Test 3**: Wed. Nov. 20
Nov. 25		Fall Break!
Dec. 2	Finish Ch9 and Review	
Dec. 9	Final Exam Week	**Final Exam:** TBD